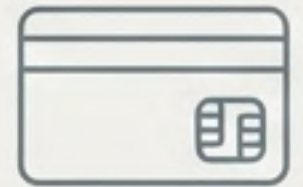




Your Algorithm May Be Illegal Under the EU AI Act 2026



A Cross-Domain Audit and Mitigation Pipeline
for High-Risk Artificial Intelligence



By 2026, preventing algorithmic bias is no longer a corporate ethical choice; it is a **strict legal requirement**. We audited classification models across four high-risk domains and actively deployed **algorithmic mitigation** to measure the exact cost of legal compliance.

The Annex III Risk Portfolio

Employment

Adult Income
dataset

EU AI Act
Annex III, Item 4

Finance

Credit Default
dataset

EU AI Act
Annex III, Item 5

Criminal Justice

COMPAS
dataset

EU AI Act
Annex III, Item 8

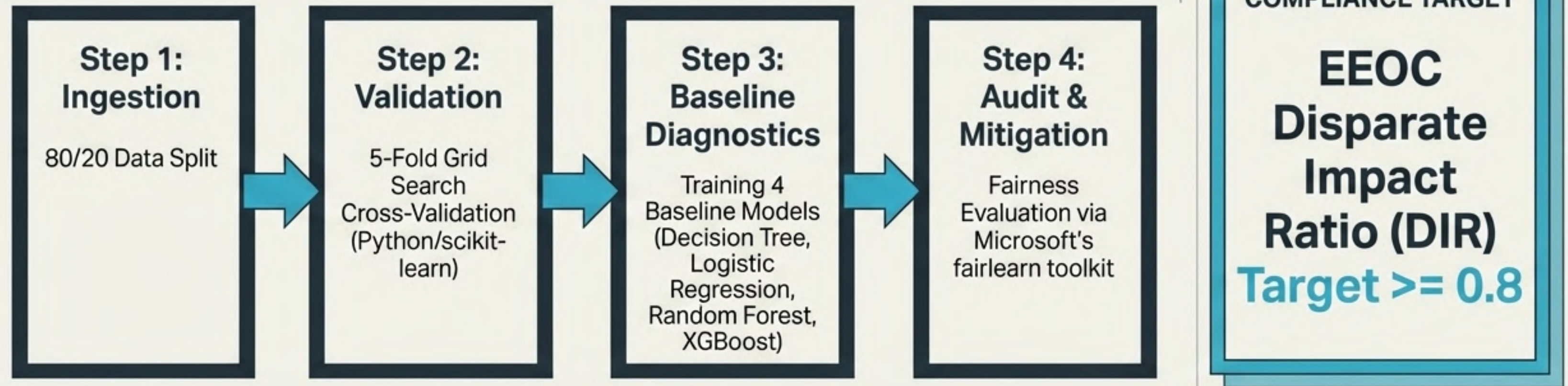
Marketing

Bank Marketing
dataset

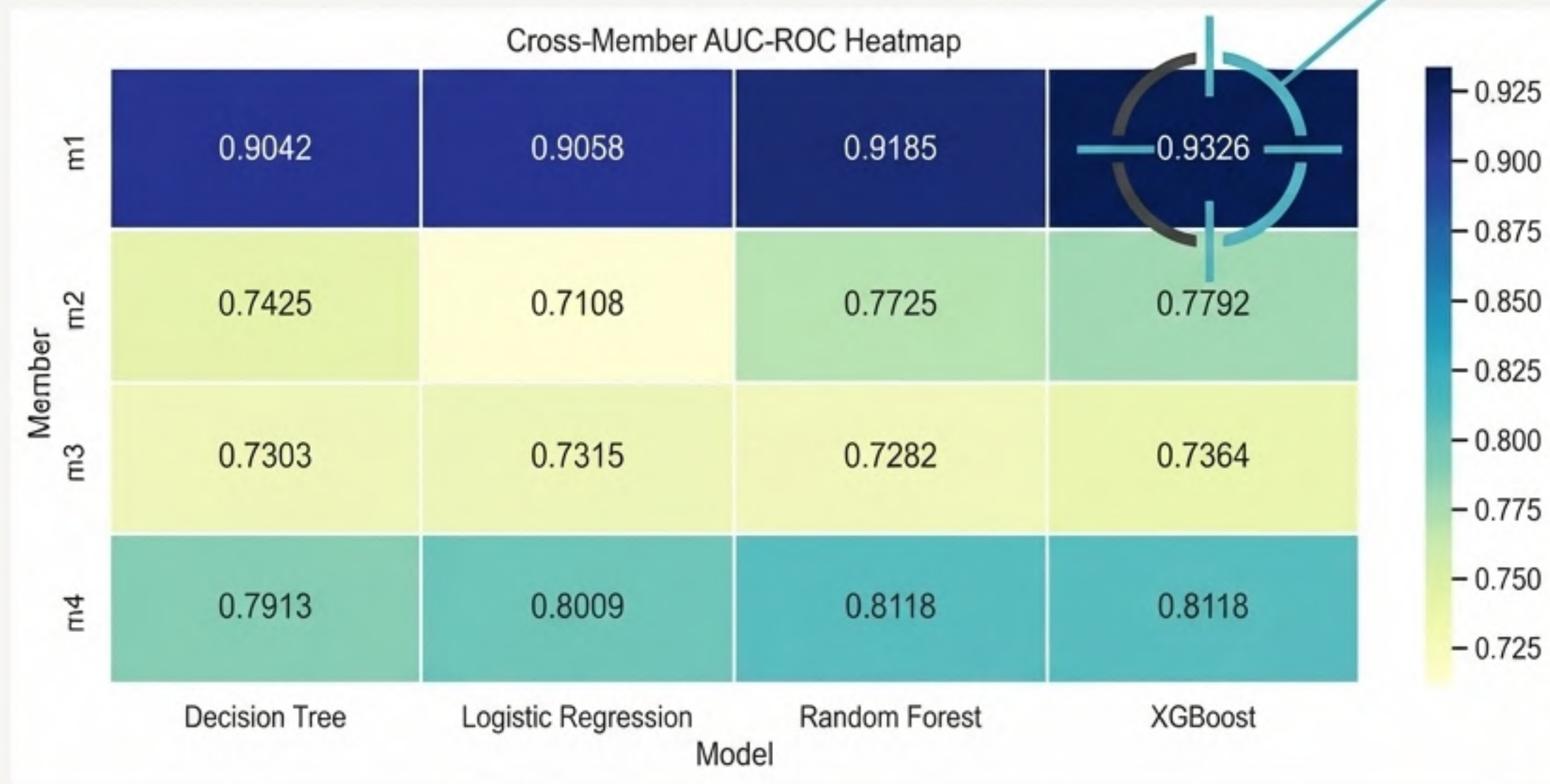
EU AI Act
Annex III, Item 5

Note: We utilize historical North American datasets as mathematical proxies to simulate bias, intentionally respecting GDPR Article 9 which heavily restricts the processing of implicit racial data in Europe.

A Standardized Diagnostic Pipeline for Algorithmic Fairness



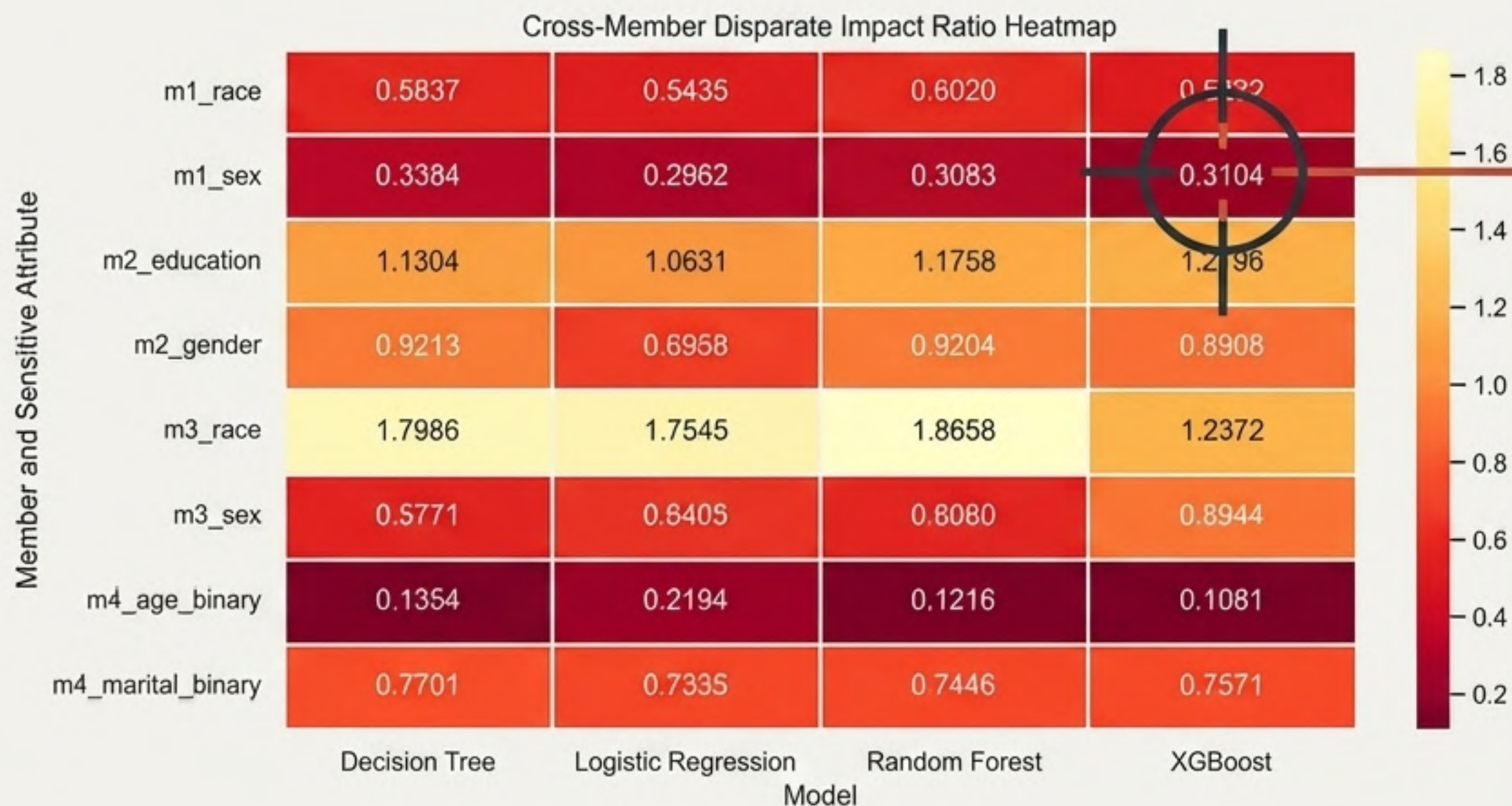
The Illusion of Perfection: Unmatched Predictive Power



0.9326

XGBoost dominates raw predictive capability across all high-risk domains. In Employment tests, AUC-ROC approaches an exceptional ~0.93. Evaluated purely on traditional accuracy metrics, this model appears ready for immediate production deployment.

The Bias Trap: Highest Performance Yields the Highest Liability



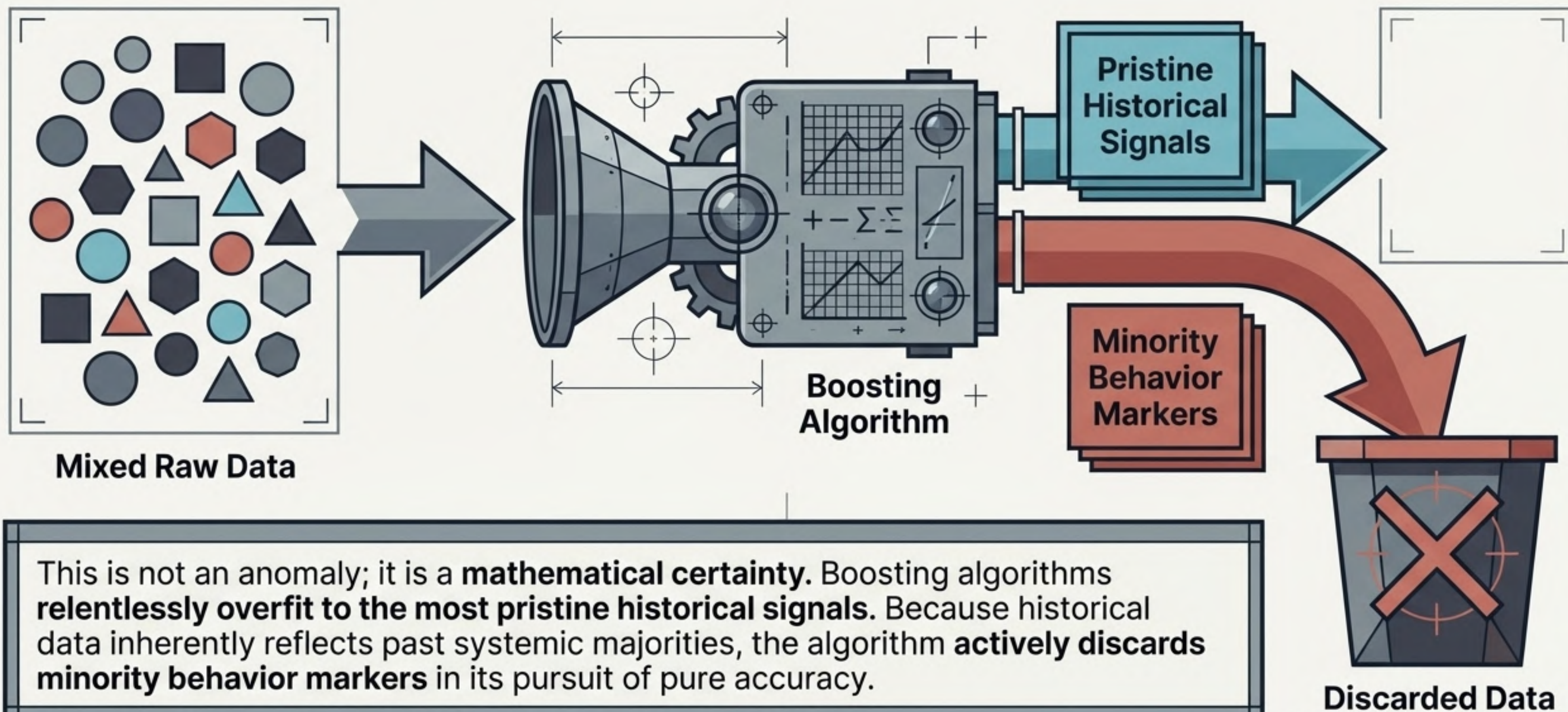
0.3104

The highest-performing model is the most legally compromised.

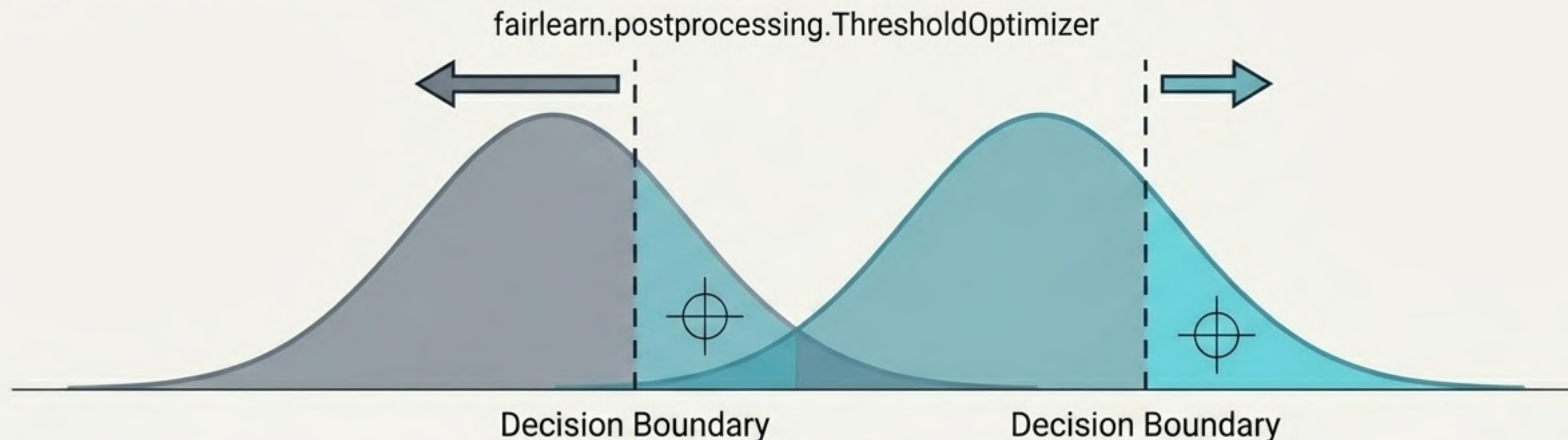
Under unmitigated XGBoost, the Female Disparate Impact Ratio in Adult Income data collapses to ~0.30—shattering the 0.80 legal threshold established by the EEOC.

Unchecked, raw optimization guarantees systemic discrimination.

The Engine of Bias: Anatomy of Overfitting

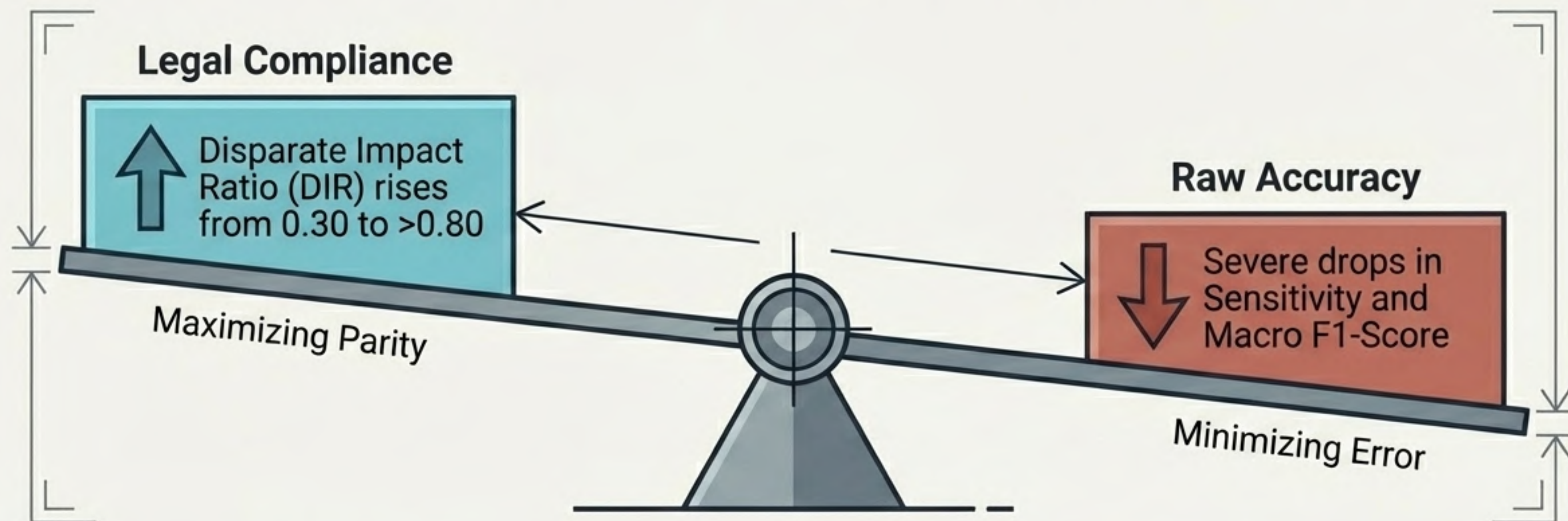


The Algorithmic Cure: Dynamic Threshold Optimization



We engineered a cure using **Post-Processing Threshold Optimization**. Rather than using a uniform cutoff, the optimizer dynamically adjusts decision boundaries exclusively for different demographic groups. This forces the XGBoost pipeline to repair its **Demographic Parity Difference** and output legally compliant **Disparate Impact Ratios** (≥ 0.8) natively.

The Impossibility Theorem: The True Cost of Compliance



Achieving strict parity mathematically destroys raw accuracy. You cannot optimize for both simultaneously. This empirically proves the '**Impossibility Theorem**': the inherent **mathematical tension** between pure predictive accuracy and strict demographic parity. High accuracy on historically skewed data guarantees bias.

The Calibration Risk: The Danger of Algorithmic Quotas



Forcing strict demographic parity introduces a dangerous side effect: the Quota Problem.

If the algorithm is mathematically forced to hit an exact demographic parity ratio, it may break its own calibration.
In Criminal Justice modeling, this can result in misclassifying legitimately high-risk defendants simply to hit a mandated numerical quota.

The GDPR Paradox: A Legal Catch-22

Post-Processing Mitigation



Requires collecting an individual's specific demographic traits at runtime to apply the correct decision boundary

GDPR Article 9



Heavily restricts the collection and processing of implicit racial or demographic data in Europe

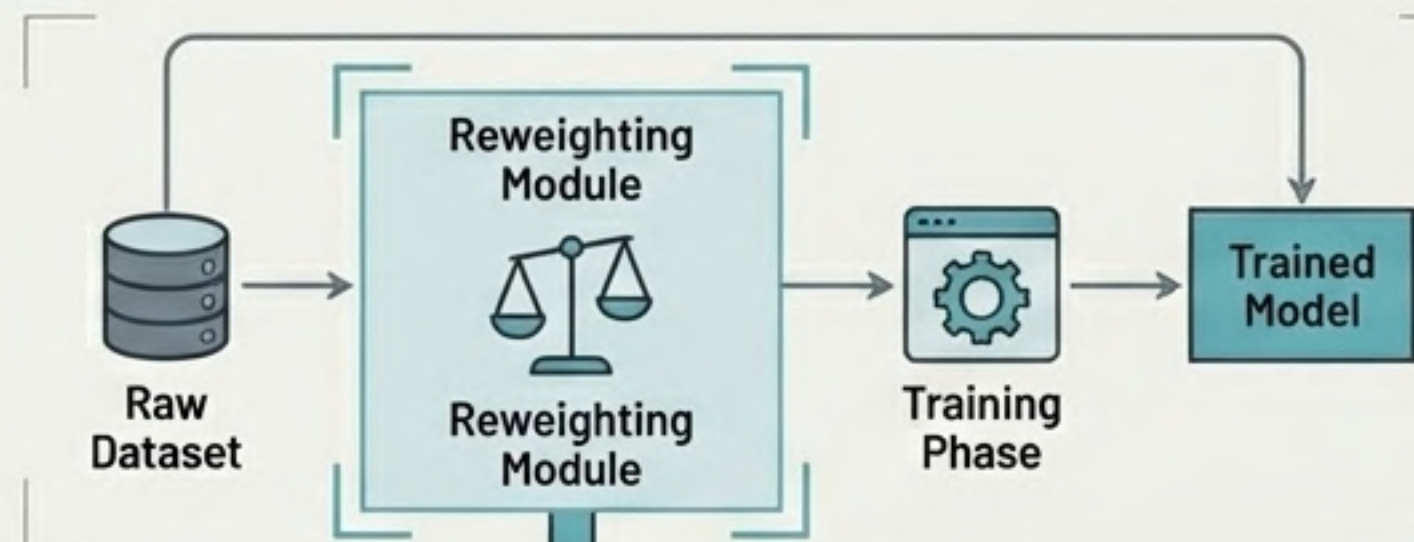
To mathematically fix the model, we must collect demographic data at evaluation runtime.
To comply with privacy law, we are forbidden from collecting that exact data.

Future Architectures for High-Risk Deployment

01.

Pivot to Pre-Processing (Reweighting)

Shift mitigation to the training phase via dataset reweighting. This repairs the training distribution upfront, completely sidestepping the GDPR paradox of requiring demographic data collection at runtime.



02.

Implement Bootstrapping

Standard Cross-Validation is insufficient for precise fairness guarantees. We must integrate bootstrapping to generate true 95% Confidence Intervals for fairness metrics, ensuring models remain legally compliant in volatile production environments.

